The “janitor” Package in R

Using **janitor** to clean, summarize, and improve data tables

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DSS 665 - R Statistical Programming

1. What is janitor?
2. Explore the functions
3. Improve the Tables

Index & Content

What is “janitor”?

“**Janitor**” provides simple tools for cleaning datasets and creating descriptive tables

Clean Data

- Standardize column names

Summarize data

- Create frequency and cross-tabulation tables

Improve tables

- Add totals and percentages

evals

Course	score	rank	ethnicity	gender	language	age	cls_perc_eval	cls_did_eval	cls_students	cls_level	cls_profs	bty_avg	pic_outfit	pic_color
1	4.7	tenure track	minority	female	english	36	55.81395	24	43	upper	single	5	not formal	color
2	4.1	tenure track	minority	female	english	36	68.8	86	125	upper	single	5	not formal	color
3	3.9	tenure track	minority	female	english	36	60.8	76	125	upper	single	5	not formal	color
4	4.8	tenure track	minority	female	english	36	62.60163	77	123	upper	single	5	not formal	color
5	4.6	tenured	not minority	male	english	59	85	17	20	upper	multiple	3	not formal	color
6	4.3	tenured	not minority	male	english	59	87.5	35	40	upper	multiple	3	not formal	color
7	2.8	tenured	not minority	male	english	59	88.63636	39	44	upper	multiple	3	not formal	color
8	4.1	tenured	not minority	male	english	51	100	55	55	upper	single	3.333	not formal	color
9	3.4	tenured	not minority	male	english	51	56.92308	111	195	upper	single	3.333	not formal	color
10	4.5	tenured	not minority	female	english	40	86.95652	40	46	upper	single	3.167	not formal	color
11	3.8	tenured	not minority	female	english	40	88.88889	24	27	upper	single	3.167	not formal	color
12	4.5	tenured	not minority	female	english	40	96	24	25	upper	multiple	3.167	not formal	color
13	4.6	tenured	not minority	female	english	40	85	17	20	upper	single	3.167	not formal	color
14	3.9	tenured	not minority	female	english	40	56	14	25	upper	single	3.167	not formal	color
15	3.9	tenured	not minority	female	english	40	88.09524	37	42	upper	single	3.167	not formal	color
16	4.3	tenured	not minority	female	english	40	90	18	20	upper	multiple	3.167	not formal	color
17	4.5	tenured	not minority	female	english	40	83.33334	15	18	upper	multiple	3.167	not formal	color
18	4.8	tenure track	not minority	female	english	31	87.5	42	48	upper	multiple	7.333	not formal	color
19	4.6	tenure track	not minority	female	english	31	90.90909	40	44	upper	multiple	7.333	not formal	color
20	4.6	tenure track	not minority	female	english	31	79.16666	38	48	upper	multiple	7.333	not formal	color

The evals Dataset

First 20 Observations

Student Evaluation Data

- 463 Observations
- 15 Variables
- Categorical and numerical variables

The Data

Gender

- Female
- Male

Rank

- Teaching
- Tenure Track
- Tenured

We'll be using these categorical variables to demonstrate **janitor**

Variables

clean_names()

```
> clean_names(evals)
# A tibble: 463 × 15
   course score rank    ethnicity gender language   age
    <dbl> <dbl> <chr>   <chr>     <chr>  <chr>    <dbl>
 1      1   4.7 tenur… minority  female english     36
 2      2   4.1 tenur… minority  female english     36
 3      3   3.9 tenur… minority  female english     36
 4      4   4.8 tenur… minority  female english     36
 5      5   4.6 tenur… not mino… male   english     59
```

Standardizes column names

tabyl()

Creates frequency tables

Shows counts and percentages

Works with one or multiple variables

Useful for categorical data

One Variable

```
> tabyl(evals, gender)
 gender   n   percent
 female 195 0.4211663
   male 268 0.5788337
```

How many observations distributed by gender?

Two Variables

```
> tabyl(evals, gender, rank)
 gender teaching tenure track tenured
 female       50           69      76
   male       52           39     177
```

How does instructor rank vary by gender?

Exploring janitor functions

Making Tables More Informative

adorn_totals()

Adds totals to a frequency table

```
> tabyl(evals, gender, rank) %>%
+   adorn_totals()
 gender teaching tenure track tenured
 female       50            69      76
   male       52            39     177
  Total      102           108     253
```

Why use it?

- Makes overall counts easier to see
- Useful with two-way tables
- Helps compare categories

adorn_percentages()

Converts counts into proportions

```
> tabyl(evals, gender, rank) %>%
+   adorn_percentages("row")
 gender  teaching tenure track   tenured
 female 0.2564103    0.3538462 0.3897436
   male 0.1940299    0.1455224 0.6604478
```

Why use it?

- Makes categories easier to compare
- “row” calculates percentages within each group
- Useful when group sizes differ

Thank you!

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